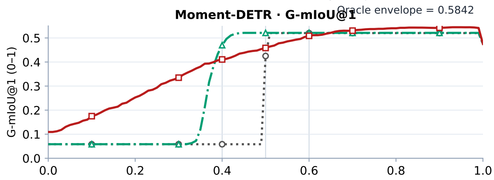
SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E9 / CAB) + GEQ

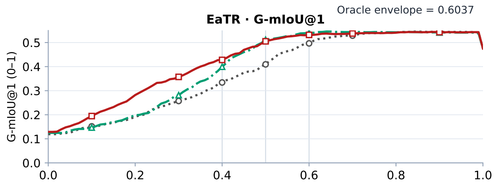
Standard-trained, fixed Val-selected best checkpoints → Standard Test (1,036 queries)

Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]



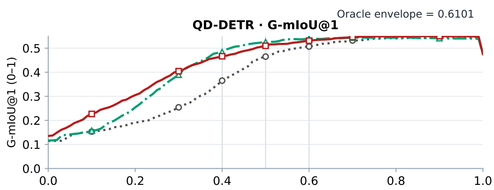
GMR baseline | mAP 6.68% · GMR + GEQ Full | mAP 5.77% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 13.61%

ΔmAP vs GMR: +6.93 pp (vs GEQ: +7.84 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +12.48 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +10.57 pp**



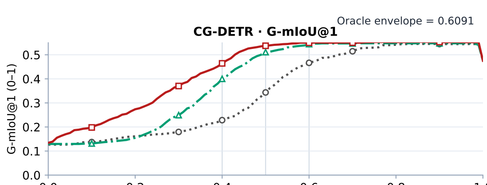
GMR baseline | mAP 16.49% · GMR + GEQ Full | mAP 17.48% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 17.23%

ΔmAP vs GMR: +0.74 pp (vs GEQ: -0.25 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +7.50 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +7.39 pp**



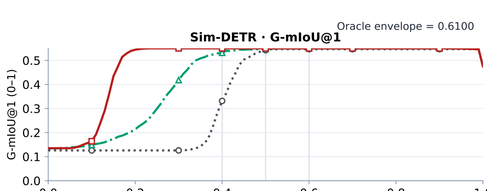
GMR baseline | mAP 14.62% · GMR + GEQ Full | mAP 14.83% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 17.65%

ΔmAP vs GMR: +3.03 pp (vs GEQ: +2.82 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +5.69 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +5.30 pp**



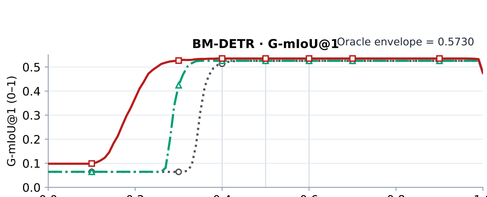
GMR baseline | mAP 18.65% · GMR + GEQ Full | mAP 18.71% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 19.34%

ΔmAP vs GMR: +0.69 pp (vs GEQ: +0.63 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +17.10 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +17.04 pp**



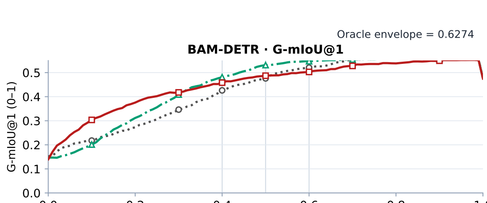
GMR baseline | mAP 20.73% · GMR + GEQ Full | mAP 22.26% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 22.45%

ΔmAP vs GMR: +1.72 pp (vs GEQ: +0.19 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +7.65 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +7.83 pp**



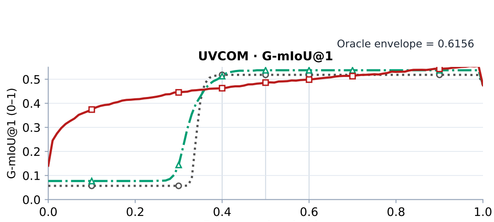
GMR baseline | mAP 4.00% · GMR + GEQ Full | mAP 4.27% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 11.87%

ΔmAP vs GMR: +7.87 pp (vs GEQ: +7.60 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +1.38 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +0.74 pp**



GMR baseline | mAP 23.25% · GMR + GEQ Full | mAP 23.00% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 23.26%

ΔmAP vs GMR: +0.01 pp (vs GEQ: +0.26 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +0.84 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +1.67 pp**



GMR baseline | mAP 7.12% · GMR + GEQ Full | mAP 8.61% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 18.51%

ΔmAP vs GMR: +11.39 pp (vs GEQ: +9.90 pp) · **ΔG1 mean vs GMR (.4/.5/.6): -3.58 pp** · **ΔG3 mean vs GMR (.4/.5/.6): -6.91 pp**

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E9) + GEQ − GMR baseline.

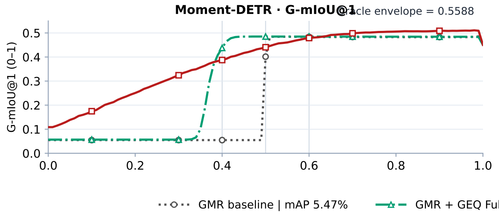
Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

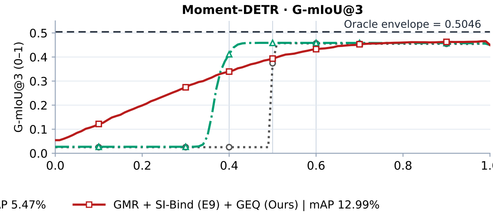
Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.

SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E9 / CAB) + GEQ

Standard-trained, fixed Val-selected best checkpoints → Full-Decodable Test (2,970 queries)

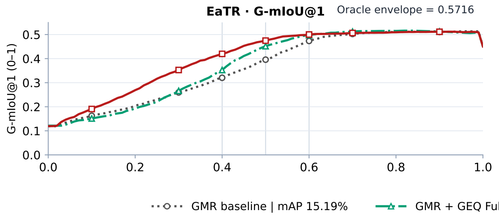
Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]

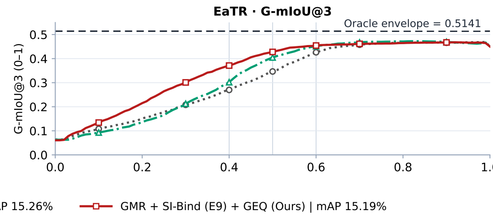




GMR baseline | mAP 5.47% · GMR + GEQ Full | mAP 5.47% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 12.99%

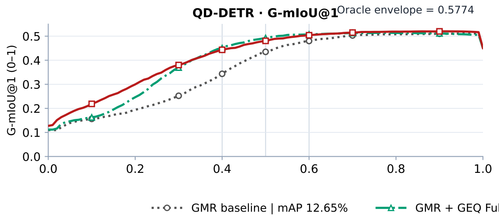
ΔmAP vs GMR: +7.52 pp (vs GEQ: +7.52 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +12.24 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +10.33 pp**

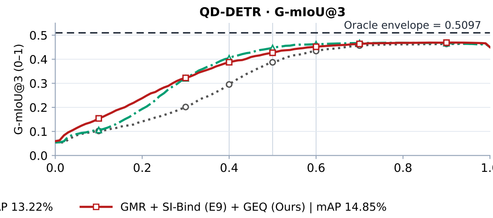




GMR baseline | mAP 15.19% · GMR + GEQ Full | mAP 15.26% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 15.19%

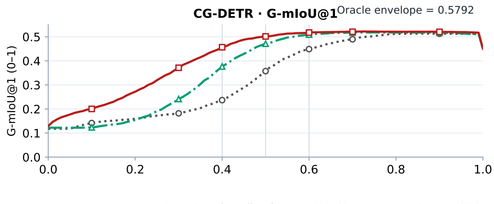
ΔmAP vs GMR: +0.00 pp (vs GEQ: -0.07 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +6.84 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +7.02 pp**

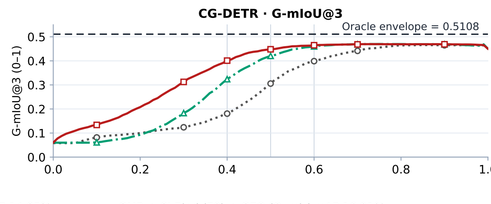




GMR baseline | mAP 12.65% · GMR + GEQ Full | mAP 13.22% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 14.85%

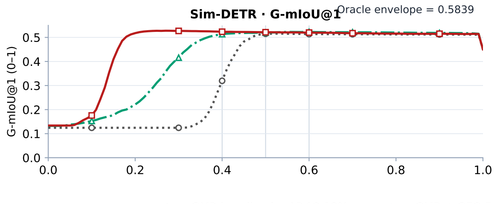
ΔmAP vs GMR: +2.20 pp (vs GEQ: +1.63 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +5.64 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +5.00 pp**

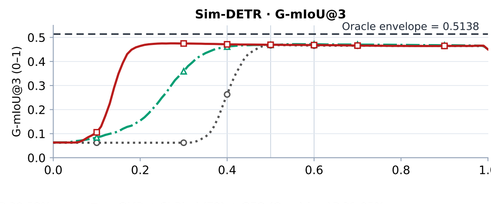




GMR baseline | mAP 14.67% · GMR + GEQ Full | mAP 14.85% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 16.91%

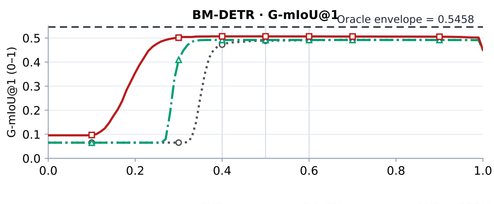
ΔmAP vs GMR: +2.24 pp (vs GEQ: +2.06 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +14.42 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +14.29 pp**

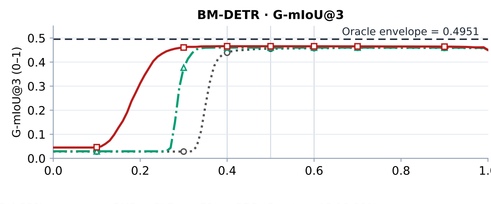




GMR baseline | mAP 19.18% · GMR + GEQ Full | mAP 20.58% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 21.05%

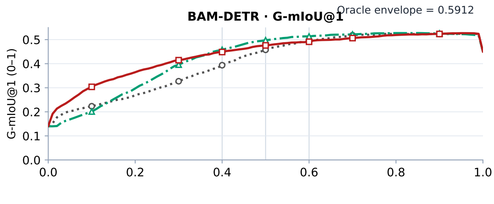
ΔmAP vs GMR: +1.87 pp (vs GEQ: +0.47 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +7.23 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +7.10 pp**

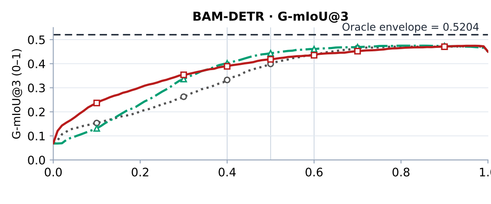




GMR baseline | mAP 3.65% · GMR + GEQ Full | mAP 4.32% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 10.91%

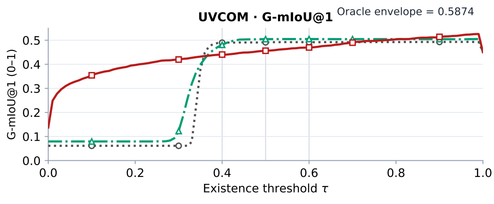
ΔmAP vs GMR: +7.26 pp (vs GEQ: +6.59 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +2.27 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +1.49 pp**

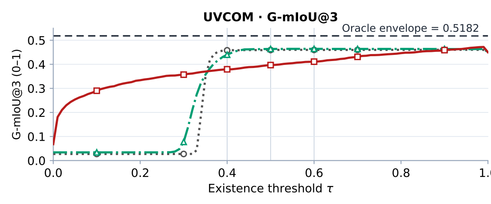




GMR baseline | mAP 20.62% · GMR + GEQ Full | mAP 20.26% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 21.43%

ΔmAP vs GMR: +0.81 pp (vs GEQ: +1.17 pp) · **ΔG1 mean vs GMR (.4/.5/.6): +2.38 pp** · **ΔG3 mean vs GMR (.4/.5/.6): +2.45 pp**





GMR baseline | mAP 6.40% · GMR + GEQ Full | mAP 8.13% · GMR + SI-Bind (E9) + GEQ (Ours) | mAP 17.37%

ΔmAP vs GMR: +10.97 pp (vs GEQ: +9.24 pp) · **ΔG1 mean vs GMR (.4/.5/.6): -3.54 pp** · **ΔG3 mean vs GMR (.4/.5/.6): -6.34 pp**

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E9) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.